Math 242, S15 Name: Answer
Quiz 3 SSN: xxx-xx-

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer.

Problem 1. (10 points) Find the particular solution of the initial value problem:

$$x\frac{dy}{dx} = 2y + x^3 \sin x, \qquad y(\pi) = 0.$$

(*First transform the differential equation into the standard form of linear first-order differential equation if it is not.*)

$$\frac{dy}{dx} = \frac{2y}{x} + x^2 \sin x$$

$$\frac{dy}{dx} - \left(\frac{2}{x}\right)y = x^2 \sin x \leftarrow \text{standard form}$$

$$P(x) = -\frac{2}{x} \quad Q(x) = x^2 \sin x$$

$$\Rightarrow \rho(x) = e^{\int(-\frac{2}{x})dx} = e^{-2\ln x} = \frac{1}{x^2} \leftarrow \text{integrating factor.}$$

Thus

$$\frac{d}{dx}\left[\left(\frac{1}{x^2}\right)y\right] = (x^2 \sin x)\cdot\left(\frac{1}{x^2}\right)$$

$$\frac{y}{x^2} = \int \sin x \, dx + C$$

$$\frac{y}{x^2} = -\cos x + C \Rightarrow y(x) = x^2(-\cos x + C)$$

$$y(\pi) = 0 \Rightarrow 0 = \pi^2(-\cos(\pi) + C)$$

$$\Rightarrow C + 1 = 0 \Rightarrow C = -1$$

Thus $y(x) = x^2(-\cos x - 1)$

$$= -x^2(\cos x + 1)$$